CODEALPHA IoT INTERNSHIP

LDR-Based Automatic Street Light

Task 2: Sensor-Based Simulation

Project Type: Arduino UNO + LDR + LED simulation
Prepared for: CodeAlpha Internship Submission

Project Report

This document explains the objective, components, circuit connections, working principle, Arduino program, code explanation, expected simulation result, and conclusion for the LDR-based automatic light control project.

Note: The threshold in the supplied program is 80. The LED turns ON when the analog reading is 80 or lower, and turns OFF when the reading is above 80. Actual readings depend on the LDR circuit and simulation light level.

1. Objective

The objective of this project is to simulate an automatic street-light control system using an LDR (Light Dependent Resistor), an Arduino UNO, and an LED. The Arduino reads the light-dependent analog value and switches the LED according to a threshold set in the program. This demonstrates a basic sensor-based automation system.

2. Components Required

Component	Purpose
Arduino UNO	Reads the sensor value and controls the LED.
LDR / Photoresistor	Changes resistance according to the light falling on it.
LED	Represents the street light in the simulation.
Resistors	Used in the LDR voltage-divider circuit and to limit LED current.
Breadboard	Provides convenient connections between components.
Jumper wires	Connect the Arduino, sensor circuit, and LED.
Tinkercad Circuits	Used to build and test the circuit virtually.

3. Circuit Connections

According to the supplied code, the LED control pin is digital pin 7 and the LDR signal is read from analog pin A0. Connect the LED (with a current-limiting resistor) to digital pin 7 and GND. Connect the LDR and a fixed resistor as a voltage divider, with the divider output connected to A0, and connect the sensor circuit to 5V and GND. Ensure the physical wiring matches these pin assignments.

Pin assignments in the code

Arduino pin	Function
D7	LED output
A0	LDR analog input
5V and GND	Power and reference for the sensor circuit

4. Working Principle

An LDR is a light-sensitive resistor: its resistance generally decreases as light intensity increases and increases in darkness. In the circuit, the LDR and a fixed resistor form a voltage divider that produces an analog voltage at A0. The Arduino converts that voltage into a reading from 0 to 1023. The program compares the reading with the threshold value 80. If the reading is 80 or below, the LED is switched ON; otherwise, the LED is switched OFF. The exact direction of readings versus light depends on how the LDR and fixed resistor are arranged in the divider.

5. Arduino Code

```
const int ledPin = 7;
const int ldrPin = A0;
```

```
void setup() {
  Serial.begin(9600);
  pinMode(ledPin, OUTPUT);
  pinMode(ldrPin, INPUT);
}

void loop() {
  int ldrStatus = analogRead(ldrPin);

  Serial.println(ldrStatus);

  if (ldrStatus <= 80) {
    digitalWrite(ledPin, HIGH);
  }
  else {
    digitalWrite(ledPin, LOW);
  }
}
```

6. Code Explanation

const int ledPin = 7; — Assigns digital pin 7 to control the LED.

const int ldrPin = A0; — Assigns analog pin A0 to read the LDR voltage.

Serial.begin(9600); — Starts serial communication at 9600 baud so readings can be viewed in the Serial Monitor.

pinMode(ledPin, OUTPUT); — Configures pin 7 as an output to control the LED.

pinMode(ldrPin, INPUT); — Configures A0 as an input. For analogRead(), explicitly setting INPUT is not strictly necessary, but it is valid.

analogRead(ldrPin) — Reads the analog input on A0 as a value from 0 to 1023.

Serial.println(ldrStatus); — Prints the current sensor reading in the Serial Monitor.

if (ldrStatus <= 80) — Checks whether the reading is less than or equal to the threshold 80.

digitalWrite(ledPin, HIGH); — Turns the LED ON when the condition is true.

digitalWrite(ledPin, LOW); — Turns the LED OFF when the reading is greater than 80.

7. Simulation Result

The Arduino repeatedly reads the LDR value and prints it to the Serial Monitor. When the value is 80 or below, the LED should be ON. When the value is above 80, the LED should be OFF. During testing, change the simulated light level and record the observed readings and LED state. Attach a screenshot of the completed Tinkercad circuit and a screenshot of the code/Serial Monitor to the final submission.

8. Conclusion

This simulation demonstrates how a sensor and microcontroller can automate a light. The LDR provides a light-dependent input, and the Arduino uses a threshold-based condition to control the LED. The project introduces analog sensor reading, digital output control, and basic automation. The same concept can be extended to a practical outdoor lighting system with suitable components and weatherproofing.

Submission Checklist

■ Completed circuit screenshot from Tinkercad

■ Arduino code file or code screenshot

■ This project explanation PDF

■ Optional: Serial Monitor screenshot showing readings and LED response